

DIGITAL FORENSIC EXAMINATION REPORT

Nitroba University Harassment Email Investigation

Case Reference: CIP-B105-CS2-C11-26-DFIT-17305

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Course/Module: DFIT – Digital Forensics & Incident Response, CIP-B105-CS2

Evidence Examined: nitroba.pcap (94,410 packets; capture window 2008-07-22 01:51:07–06:13:47 UTC)

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CLASSIFICATION: Training / Laboratory Exercise Not a Live Legal Proceeding

For educational use only. Evidence items are course-supplied network-capture artifacts.

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1. Executive Summary

This examination analyzed a packet capture (nitroba.pcap) supplied as evidence in a harassment complaint filed by instructor Lily Tuckrige of Chemistry 109, who reported receiving a threatening electronic message warning her to stop teaching. Analysis of the capture identified the sending device by IP address (192.168.15.4) and Ethernet MAC address (00:17:f2:e2:c0:ce), and recovered the verbatim content of the threatening submission made via a web form to www.willselfdestruct.com at 2008-07-22 06:04:24 UTC.

Two distinct identity linked artifacts were recovered from the same device within the capture window an authenticated Facebook session (beth@bethr.org) and an authenticated Gmail/Google Calendar session (jcoachj@gmail.com), the latter occurring approximately 3.5 minutes before the threatening submission. Comparison against the Chemistry 109 class roster identifies a plausible name match for the Gmail identity (“Johnny Coach”), though this is a naming pattern inference rather than a verified account ownership lookup.

The evidence is assessed as strongly supporting device- and session-level attribution, with an important limitation: the network was an unsecured, shared connection, and a second, unresolved identity artifact was recovered from the same device. The calibrated final conclusion, together with the confidence levels attached to each element of the attribution, is set out in Section 7 and Section 8.

2. Authority, Scope and Investigative Questions

2.1 Authority

This examination was performed as a supervised laboratory exercise under course module CIP-B105-CS2 (Digital Forensics and Incident Response), the second case in the same coursework series as the preceding CIP-B105-CS1 examination. Evidence was supplied by the course instructor for controlled analysis. This document follows professional report structure and forensic method for training purposes; it does not constitute a report prepared under an independent legal or investigative mandate.

2.2 Scope

This examination is limited to forensic analysis of the single supplied network capture file, nitroba.pcap. No physical devices, live systems, or other evidence sources were available or examined. Analysis is restricted to information observable within the packet capture itself header metadata (IP, MAC, timestamps) and payload content transmitted in cleartext (HTTP).

2.3 Investigative Questions

- Was a threatening message actually transmitted over this network, and what did it say?
- Which IP address and device (MAC address) sent it, and to what destination?
- Can the sending device be linked to a specific individual's online identity?
- Does that identity correspond to a name on the Chemistry 109 class roster?
- What is the confidence level of this attribution, and what alternative explanations must be considered given a shared, unsecured network?

3. Evidence Inventory and Integrity Verification

3.1 Evidence Inventory

Item	Description	Format	In Scope?
nitroba.pcap	Main case evidence 94,410 packets	libpcap, microsecond timestamps, v2.4	Yes
mycapture.pcap	Unrelated practice capture (separate tshark introduction exercise)	pcapng v1.0	No
mycapture10_10_81.75.pcap	Unrelated practice capture	pcapng v1.0	No
mycapture10_10_81_0.pcap	Unrelated practice capture	pcapng v1.0	No
ubalt.pcap	Unrelated practice capture (ubalt.edu subnet)	pcapng v1.0	No

The four excluded files were reviewed and confirmed to belong to a separate training exercise (different IP subnet, different scenario, pcapng rather than legacy pcap format); none are part of this case's evidentiary chain.

3.2 Hash Verification

File	MD5	SHA-256
nitroba.pcap	9981827f11968773ff815e39f5458ec8	2b77a9eafc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb

Working copy verification a working copy was created (cp evidence/nitroba.pcap working/nitroba.pcap) and rehashed to confirm an exact match prior to any analysis. All queries in this report were run against the working copy, not the original.

4. Tools, Versions and Methodology

4.1 Environment and Tools

Tool	Purpose
tshark (Wireshark CLI)	Frame level filtering, field extraction and HTTP payload decoding
md5sum / sha256sum	Evidence and working copy integrity verification
grep / sed	Pattern extraction and normalization of tshark text output
Platform	Kali Linux (VM), VirtualBox shared folder used for evidence transfer

4.2 Methodology

- Preserve evidence copy into a dedicated case folder; hash the original and the working copy.
- Establish capture scope packet count, time range, protocol hierarchy.
- Locate the alleged malicious activity by content-based filtering (a display filter on the known destination site name).
- Confirm the web service and client IP by cross-referencing the request and its response (page-title match).
- Extract and verify the transmitted payload (HTTP POST body) using at least two independent tshark extraction methods.
- Identify the client device's MAC address from the same frame as the POST.
- Search all traffic from that MAC address for identity-bearing artifacts (cookies, authenticated session parameters).
- Cross-reference any recovered identity against the class roster.
- Normalize all timestamps to UTC; note any secondary timezone shown for readability only.
- Assess alternative explanations before drawing a final, calibrated conclusion.

5. Detailed Findings and Evidential Interpretation

5.1 Capture Overview

Property	Value
Total packets	94,410
Capture start (UTC)	2008-07-22 01:51:07
Capture end (UTC)	2008-07-22 06:13:47
Duration	Approximately 4 hours 22 minutes
Format	libpcap, microsecond timestamps, little-endian, v2.4, Ethernet

5.2 Website Flow Reconstruction

Traffic referencing “willselfdestruct” was filtered from the full capture. The service was confirmed at IP 69.25.94.22 by matching the page title returned in the HTTP response body. The client was confirmed at IP 192.168.15.4 from the corresponding request side of the same TCP conversation.

Frame	Time (UTC)	Direction	Method / Response	URI
82935	06:03:43	client → server	GET	/secure/submit
83600	06:04:24	client → server	POST (Content-Length: 188)	/secure/submit
83603	06:04:24	server → client	302 redirect	→ /secure/success
83613	06:04:24	client → server	GET	/secure/success

5.3 Recovered POST Content

The full form body was extracted independently via (a) a direct hex/ASCII dump filtered on the destination string, and (b) urlencoded form field decoding both methods returned identical content, confirmed as:

to=lilytuckrige@yahoo.com

from=(blank)

subject="you can't find us"

message="and you can't hide from us. Stop teaching. Start running."

type=0

ttl=30

This directly confirms the complainant’s allegation a threatening message was submitted through this web service, addressed to her own e-mail account, within the captured window.

5.4 Client Device Identification

MAC address 00:17:f2:e2:c0:ce was extracted from the Ethernet source field of frame 83600 (the POST frame itself) and confirmed consistent across every frame in the same TCP stream.

An IP address alone was not treated as sufficient identification, since the network in question is a shared residential connection potentially used by multiple occupants (see Section 7). The MAC address narrows attribution to a specific network interface or device, rather than merely a shared IP.

5.5 Identity Artifacts Recovered From the Same Device

Identity String	Occurrences	Source Artifact
beth@bethr.org	137	Facebook login_x cookie (PHP-serialized session data)
jcoachj@gmail.com	113	Google Calendar/Gmail authenticated session parameter (gausr=)
lilytuckrige@yahoo.com	2	Appears only within the POST body itself the complainant’s own address, not a suspect identity
cambridgecommons@hotmail.com	2	Unrelated background traffic
the_whole_world_is_watching@nitroba.org	1	Unrelated background traffic

Timing correlation the Gmail/Calendar session under jcoachj@gmail.com was active from 06:00:44 to 06:01:02 UTC approximately 3.5 minutes before the threatening POST at 06:04:24 UTC, on the same MAC address. This close temporal proximity strengthens the association between this identity and the device used to submit the threat, though it does not by itself prove that the same person performed both actions.

5.6 Roster Comparison

Chemistry 109 roster (instructor: Lily Tuckrige): Amy Smith, Burt Greedom, Tuck Gorge, Ava Book, Johnny Coach, Jeremy Ledvkin, Nancy Colburne, Tamara Perkins, Esther Pringle, Asar Misrad, Jenny Kant.

Finding: the identity string “jcoachj@gmail.com” is consistent with the roster name “Johnny Coach” by initials/surname pattern. “beth@bethr.org” does not correspond to any name on the roster.

This name match is treated as an inference drawn from naming convention and initials, not as a verified account-ownership record — no subpoenaed or account-provider verification was available for this examination. It is reported at moderate confidence rather than certainty in Section 7.3.

6. Integrated Timeline (UTC)

Time (UTC)	Event	Source
01:51:07	Capture begins	Frame 1
06:00:44–06:01:02	Authenticated Gmail/Calendar session as jcoachj@gmail.com, same MAC address	Gmail/Calendar identity extraction (queries)
06:03:43	GET /secure/submit (form page requested)	Frame 82935
06:04:24	POST /secure/submit — threatening message submitted	Frame 83600
06:04:24	302 redirect to /secure/success	Frame 83603
06:04:24	GET /secure/success (confirmation page loaded)	Frame 83613
06:13:47	Capture ends	Last frame

The Facebook session (beth@bethr.org) is referenced by its associated frame range in Appendix B rather than as a separate timeline entry: within this report its evidentiary role is as a secondary, non-primary identity artifact (Sections 5.5 and 7.1), whereas the timeline above is organized around the primary event chain the threatening submission itself and the identity session most closely preceding it.

Timezone note: all times reported above are UTC, per the capture's own metadata. Local dormitory time would be EDT (UTC−4); for example, the 06:04:24 UTC submission corresponds to 02:04:24 EDT, shown here only for cross-reference.

7. Alternative Explanations, Limitations and Confidence Assessment

7.1 Alternative Explanations

- The dormitory network was shared among multiple occupants and was unsecured (no password) any of them, or a visitor or neighbor within range, could have used this connection at the relevant time.
- A MAC address identifies a device's network interface, not conclusively its owner or operator at a specific moment a borrowed, unattended, or communally-used laptop is not excluded by packet evidence alone.
- Two distinct, non-matching identity artifacts (beth@bethr.org and jcoachj@gmail.com) were recovered from the same MAC address. This could indicate: (a) one person owning both accounts, (b) a shared household device used by multiple people, or (c) two separate, temporally close but unrelated sessions. The evidence does not distinguish between these possibilities.

- The Gmail-to-roster name match (“jcoachj” → “Johnny Coach”) is a plausible inference, not a confirmed account-registration lookup.

7.2 Limitations of This Examination

- Only one evidence source (a single packet capture) was available; no host-based, physical-device, or account-provider records were obtainable to corroborate identity.
- Packet-capture evidence proves what was transmitted over the network, not who was physically operating the keyboard at that moment.
- The capture window (approximately 4 hours 22 minutes) may not include the complainant’s own retrieval of the message or subsequent events, limiting what can be reconstructed end-to-end.
- Facebook and Gmail identity artifacts were recovered via cleartext cookies and parameters as transmitted in 2008; this method would not work against modern encrypted (HTTPS) traffic the technique is date- and technology-specific to this evidence, not a general-purpose approach.
- A full chronological shell/command activity log, of the kind provided as Appendix A in the companion CIP-B105-CS1 report, was not available for this examination. Appendix D reproduces the specific commands used and is sufficient to independently reproduce every finding in Sections 5 and 6; exact tool version numbers (e.g., the installed tshark build) were likewise not separately logged.

7.3 Confidence Assessment

High confidence that a threatening message matching the complaint was transmitted from IP 192.168.15.4 / MAC 00:17:f2:e2:c0:ce at 2008-07-22 06:04:24 UTC — this is directly observed in the capture. Moderate confidence that this device was, at the relevant time, associated with a user authenticated as jcoachj@gmail.com, based on the temporal proximity of an authenticated session on the same device. Lower confidence in translating that e-mail identity to the specific individual “Johnny Coach” without independent account-ownership verification. Confidence in sole, exclusive attribution to one individual is further reduced by the shared/unsecured network and the unresolved second identity artifact.

8. Conclusion

The packet capture evidence directly substantiates the complainant’s allegation that a threatening electronic message was transmitted, and recovers its exact content. The evidence strongly supports that the transmitting device can be identified by IP and MAC address, and is consistent with though it does not conclusively prove, beyond the alternative explanations discussed in Section 7 that this device was being used at the time by an individual identifiable as “Johnny Coach” on the Chemistry 109 roster.

It is recommended that any further action, disciplinary or otherwise, be supported by additional corroborating evidence such as direct account provider records, device possession records, or an admission before identity attribution is treated as conclusively established on this packet evidence alone.

9. Appendices

Appendix A Examination Activity Log

A command-by-command shell activity log equivalent to Appendix A of the companion CIP-B105-CS1 report was not included among the materials reviewed for this examination. The commands actually used to produce every finding in this report are reproduced in full, with their purpose, in Appendix D, and are sufficient on their own to independently reproduce each result in Sections 5 and 6.

Appendix B Evidence Register

Evidence ID	Source	Reference	Purpose
EV-01	nitroba.pcap	Frames 82935, 83600, 83603, 83613	Website flow / POST content reconstruction
EV-02	nitroba.pcap	Frame 83600 (eth.src)	Client MAC identification
EV-03	nitroba.pcap	Facebook login_x cookie frames	Identity artifact beth@bethr.org
EV-04	nitroba.pcap	Gmail/Calendar gausr= frames	Identity artifact jcoachj@gmail.com

Appendix C Hash Verification

File	MD5	SHA-256
nitroba.pcap	9981827f11968773ff815e39f5458ec8	2b77a9eafc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb

Appendix D Commands and Queries Used

All commands below were executed against the working copy (working/nitroba.pcap), not the original evidence file.

1. Locate the threatening submission by destination-site content filter:

```
tshark -r working/nitroba.pcap -Y 'http contains "willselfdestruct" \
-T fields -e frame.number -e frame.time -e ip.src -e ip.dst \
-e http.request.uri -e http.host
```

2. Extract the POST body fields from the identified frame:

```
tshark -r working/nitroba.pcap -Y 'frame.number==83600' \
-T fields -e urlencoded-form.key -e urlencoded-form.value
```

3. Identify the client device's MAC address from the same frame:

```
tshark -r working/nitroba.pcap -Y 'frame.number==83600' \  
-T fields -e eth.src -e eth.dst
```

4. Search all traffic from that MAC address for identity-bearing artifacts:

```
tshark -r working/nitroba.pcap -Y 'eth.addr==00:17:f2:e2:c0:ce and http' \  
-T fields -e frame.number -e frame.time -e http.host -e http.cookie \  
| grep -oE '[A-Za-z0-9._%+~]+(%40|@)[A-Za-z0-9.-]+\.[A-Za-z]{2,}'
```

End of report.